The Starting Point of Complex Number

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Abstract: $\sqrt{-1}$ plays a vital in complex numbers. Also, there are no complex numbers without $i = \sqrt{-1}$. This paper discusses the complex numbers and its usage in the number system and also provides a lemma.

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1. Introduction

The square root of the positive one is ±1, but the negative one has no real square roots. $\sqrt{-1}$ is named as an imaginary number, i.e. $i = \sqrt{-1}$.

2. Imaginary Number

When we try to find the solution for the algebraic equation $x^2 + 1 = 0$, the complex number came for the usage in the number systems. $\sqrt{-1}$ plays a vital in complex numbers. Also, there are no complex numbers without $\sqrt{-1}$.

Here, $i = \sqrt{-1}$ denote the imaginary number. Also, $i^2 = -1$. The equation $x^2 + 1 = 0$ is true if and only if $x^2 = i^2$, where $i^2 = -1$ and $i = \sqrt{-1}$.

$a + ib$ is the general form of a complex number, where a and b are real numbers.

Lemma: $\sqrt{-1} = \dfrac{\pm 1}{\sqrt{-1}}$.

Proof.

$\sqrt{-1}\sqrt{-1} = \sqrt{-1}\sqrt{-1}$ is obviously true.

Then, $\sqrt{-1}\sqrt{-1} = \sqrt{-1}\sqrt{-1} \Rightarrow \dfrac{\sqrt{-1}\sqrt{-1}}{\sqrt{-1}\sqrt{-1}} = 1$ where $\sqrt{-1}\sqrt{-1} = \pm 1 \Rightarrow \left(\sqrt{-1}\right)^2 = \pm 1$.

$i.e.$ since $\sqrt{-1}\sqrt{-1} = \pm 1, \dfrac{\sqrt{-1}\sqrt{-1}}{\sqrt{-1}\sqrt{-1}} = \dfrac{-1}{-1} = \dfrac{1}{1} = 1$ is obviously true.

First, let us prove that $\sqrt{-1} = \dfrac{-1}{\sqrt{-1}}$.

$\sqrt{-1} = \dfrac{-1}{\sqrt{-1}}. \Rightarrow \sqrt{-1} = \dfrac{\sqrt{-1}\sqrt{-1}}{\sqrt{-1}}$, which is true for $\sqrt{-1}\sqrt{-1} = -1$. (1)

Next, let us prove that $\sqrt{-1} = \frac{+1}{\sqrt{-1}}$.

$\sqrt{-1} = \frac{+1}{\sqrt{-1}}. \Rightarrow \sqrt{-1} = \frac{\sqrt{-1}\sqrt{-1}}{\sqrt{-1}}$, which is true for $\sqrt{-1}\sqrt{-1} = +1$. (2)

From (1)and (2), we conclude that $\sqrt{-1} = \frac{\pm 1}{\sqrt{-1}}, i.e. \sqrt{-1} = \frac{1}{\sqrt{-1}}$ (OR) $\sqrt{-1} = \frac{-1}{\sqrt{-1}}$.

Hence, the lemma is proved.

3. Conclusion

When we try to find the solution for the algebraic equation $x^2 + 1 = 0$, the complex number came for the usage in the number systems. $\sqrt{-1}$ plays a vital in complex number system. Also, a lemma is introduced in the article.

References

[1] Wikipedia. https://en.wikipedia.org/wiki/Complex_number.